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Comparative analysis of machine learning models for malaria detection using validated synthetic data: a cost-sensitive approach with clinical domain knowledge integration

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Malaria remains a critical global health challenge requiring innovative diagnostic approaches. Machine learning offers promising solutions for automated detection, but systematic algorithm comparison using clinically validated data remains limited. We systematically compared five machine learning models—Naive Bayes, Logistic Regression, Random Forest, XGBoost, and Enhanced Bayesian Logistic Regression—for malaria detection using a rigorously validated synthetic dataset ($N = 10,100$) representing Sub-Saharan African epidemiological conditions. The dataset achieved 87% representativeness against published clinical benchmarks. Cost-sensitive threshold optimization prioritized clinical sensitivity ($C_{FN} = 15$, $C_{FP} = 3$). Performance evaluation employed comprehensive metrics with bootstrap confidence intervals, statistical significance testing, and clinical cost analysis. Five machine learning algorithms were evaluated using comprehensive statistical validation including bootstrap confidence intervals and significance testing. XGBoost achieved optimal performance with highest AUC (0.956, 95% CI: 0.952–0.961) and competitive clinical cost (5,496), representing 2.8% improvement over Random Forest. Enhanced Bayesian Logistic Regression incorporating clinical domain knowledge achieved comparable performance (AUC: 0.954, 95% CI: 0.950–0.959) with interpretable clinical coefficients. McNemar's test revealed statistically significant classification differences between XGBoost and Random Forest ($\chi^2 = 1508.6$, $p < 0.001$), while Friedman test indicated no significant overall ranking differences across models ($\chi^2 = 4.0$, $df = 4$, $p = 0.406$). Complete analysis pipeline with 2-step reproduction process and synthetic datasets are publicly available for independent validation and collaborative research extension. XGBoost demonstrates optimal balance of accuracy and cost-effectiveness for malaria screening applications. The systematic validation framework and cost-sensitive optimization provide practical guidance for clinical implementation. While synthetic data enables controlled algorithm comparison, real-world clinical validation remains essential before deployment.

Keywords Machine learning, Malaria detection, Synthetic data validation, Cost-sensitive classification, Clinical decision support, XGBoost, Bayesian neural networks

Malaria continues to pose a significant global health challenge, with the World Health Organization reporting 247 million cases and approximately 619,000 deaths in 2021, predominantly affecting children under five in Sub-Saharan Africa[1]. Early and accurate diagnosis remains critical for effective treatment and disease control, yet traditional diagnostic methods face substantial limitations in resource-constrained settings, including dependence on trained microscopists, inconsistent quality control, and limited diagnostic throughput[2,3].

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Machine learning approaches offer promising alternatives for automated malaria detection, potentially enhancing diagnostic accuracy, speed, and accessibility[4,5]. These computational methods can identify complex patterns in patient data that may not be immediately apparent to human observers, enabling earlier intervention and improved clinical outcomes. Recent studies have demonstrated encouraging results with various machine learning algorithms for malaria diagnosis using clinical, environmental, and imaging data[3,5].

The use of synthetic data in developing machine learning models for healthcare applications provides several critical advantages: it mitigates privacy concerns associated with sensitive patient information, enables controlled experimental conditions for systematic algorithm assessment, and allows generation of balanced datasets with known ground truth for fair model comparison[6,7]. While synthetic data cannot entirely replace real clinical data for final validation, it provides an invaluable testing ground for initial model development and comparative analysis.

However, recent systematic reviews have identified several methodological gaps in machine learning approaches for malaria detection[4,5]. Specifically, many studies lack rigorous synthetic data validation against established epidemiological benchmarks, with limited assessment of data representativeness[6]. Furthermore, insufficient statistical frameworks for performance evaluation have been noted, including inadequate confidence interval reporting and limited significance testing[3]. Recent reviews also highlight that machine learning applications in healthcare often prioritize accuracy metrics without considering clinical consequences and deployment feasibility[5], while interpretability considerations crucial for healthcare implementation remain inadequately addressed[4]. These methodological limitations may hinder the translation of research findings into practical diagnostic tools for resource-constrained healthcare settings[1].

This study addresses these limitations through a comprehensive comparative analysis of five machine learning models—Naive Bayes, Logistic Regression, Random Forest, XGBoost, and Enhanced Bayesian Logistic Regression—using a rigorously validated synthetic dataset representing Sub-Saharan African malaria epidemiology. Our approach incorporates several methodological innovations: systematic validation against published clinical benchmarks achieving 87% representativeness, cost-sensitive threshold optimization prioritizing clinical sensitivity over specificity, enhanced Bayesian methodology integrating clinical domain knowledge, and comprehensive statistical assessment with bootstrap confidence intervals and significance testing.

By incorporating cost-sensitive optimization that prioritizes sensitivity (minimizing false negatives) over specificity (minimizing false positives), this study simulates real-world clinical priorities where missed malaria diagnoses carry more severe consequences than unnecessary treatment. The findings provide evidence-based guidance for algorithm selection, clinical implementation strategies, and future research directions in automated malaria diagnosis.

Materials and methods

Synthetic data generation and validation methodology

Epidemiological foundation and parameter justification

The synthetic dataset was systematically generated to emulate realistic malaria transmission patterns in rural Sub-Saharan African communities, based on established epidemiological literature and WHO surveillance data. All parameters were derived from peer-reviewed clinical studies and validated against published epidemiological benchmarks to ensure clinical relevance and methodological rigor.

Sample size justification The synthetic population size of $N = 10,100$ was selected to represent a realistic healthcare catchment area for rural health centers in the Gambella region, Ethiopia. This choice reflects the demographic and healthcare infrastructure reality of rural Sub-Saharan Africa, where health centers typically serve 15,000–25,000 people through networks of 5–7 health posts, each covering 3,000–5,000 individuals[8,9].

Gambella region, with a total population of approximately 525,000 (75% rural)[10,11], provides the demographic foundation for this sample size choice. The 10,100 individuals represent approximately 3.6% of a typical district population or the aggregated catchment reflecting documented healthcare service areas in rural Ethiopia[12,13]. This population size aligns with healthcare catchment areas where the majority of health centers serve 2,000–5,000 people within walking distance[8].

From a statistical perspective, the sample size meets established guidelines for machine learning diagnostic studies ($\geq 10,000$ samples for robust performance assessment) while providing sufficient positive cases ($n = 1,969$, 19.5% prevalence) and negative cases ($n = 8,131$) for accurate diagnostic performance evaluation[14,15]. The sample represents the realistic patient volume that would utilize automated diagnostic tools in rural health centers, supporting both scientific validity and practical implementation relevance.

Population characteristics The synthetic population ($N = 10,100$) was designed to represent a high-transmission malaria-endemic community with 20% prevalence, consistent with WHO estimates for rural Sub-Saharan Africa[16]. This prevalence level reflects areas with perennial transmission and limited vector control interventions, typical of resource-constrained settings where automated diagnostic tools would have greatest impact.

Demographic parameters Age distribution followed UN Population Division data for Sub-Saharan Africa[17], with mean age 22.8 years (SD = 16.5), reflecting the young population structure characteristic of these regions. Age ranges were constrained to 0.5–85 years to represent realistic clinical populations seeking malaria diagnosis. Demographic patterns were validated against Tanzania Demographic and Health Survey 2015-16[18] to ensure regional representativeness.

Clinical parameter derivation and literature validation

Clinical manifestation probabilities were systematically derived from large-scale clinical studies and systematic reviews in malaria-endemic regions. Fever prevalence was established at 85% in malaria-positive cases, based

on the comprehensive study by Roucher et al.[19] involving 2,847 microscopy-confirmed cases in Senegal. Chills occurrence was documented at 78% prevalence, derived from the clinical cohort study conducted by Kahigwa et al.[20] in Tanzania, while fatigue manifestation was observed in 82% of cases, based on the community survey by Mwangi et al.[21] in coastal Kenya involving 1,844 participants.

For malaria-negative individuals, background symptom rates reflected the prevalence of other febrile illnesses commonly encountered in Sub-Saharan Africa. These baseline rates were established as fever occurring in 25% of cases, chills in 15%, and fatigue in 35%, based on comprehensive community health surveys conducted by Snow et al.[22]. These differential symptom probabilities ensure realistic representation of the diagnostic challenge in endemic areas where multiple febrile illnesses present with overlapping clinical features.

Symptom probabilities were carefully adjusted to incorporate age-specific immune responses consistently observed in endemic populations[2,6]. Children under five years demonstrated 15% higher symptom probability, reflecting their limited acquired immunity and increased vulnerability to severe manifestations. Elderly individuals over 65 years showed 10% higher symptom probability, representing the waning immunity that occurs with advanced age. Adults between 5 and 65 years exhibited 5% lower symptom probability, reflecting the partial acquired immunity that develops from repeated exposure to malaria parasites in endemic settings.

Environmental factor modeling and climate integration

Temperature Atmospheric temperature (°C) was modeled using normal distribution ($\mu = 26.5, \sigma = 4.2$) based on meteorological data from malaria-endemic regions[6]. A positive correlation with malaria status (+0.8°C mean shift for positive cases) reflected optimal transmission temperatures (20–30°C range) established by Paaijmans et al.[23].

Rainfall Monthly rainfall (mm) followed gamma distribution (shape=1.8, scale=12.5) reflecting typical seasonal patterns in Sub-Saharan Africa[23,24]. This distribution captures the right-skewed nature of precipitation data and correlation with increased vector breeding sites and malaria transmission intensity.

Feature interaction modeling and biological realism

Unlike simple random generation approaches, the synthetic data incorporated clinically relevant interactions that reflect the complex biological and epidemiological relationships observed in real-world malaria transmission. Age-symptom interactions were modeled to capture the immune response maturation that affects symptom expression patterns across different age groups, ensuring that younger and older populations exhibited appropriate vulnerability profiles. Environmental-disease interactions incorporated the established relationships between temperature and rainfall patterns and malaria transmission probability, reflecting the well-documented influence of climatic factors on vector breeding and parasite development.

Symptom clustering patterns were implemented to ensure realistic co-occurrence of fever, chills, and fatigue, avoiding the unrealistic independence that would result from purely random generation. Additionally, seasonal effects were incorporated to model how environmental factors influence both symptom severity and disease progression, capturing the temporal dynamics that characterize malaria epidemiology in endemic regions. These interaction patterns ensure that the synthetic data maintains biological plausibility while providing a controlled framework for systematic algorithm comparison.

Data generation algorithm

The Monte Carlo simulation approach employed the following systematic process:

For each individual *i* in population $N = 10,100$:

$$M_i \sim \text{Bernoulli}(0.20) \tag{1}$$

$$\text{Age}_i \sim \mathcal{N}(22.8, 16.5^2) \text{ [constrained to 0.5–85 years]} \tag{2}$$

$$\text{Temp}_i \sim \mathcal{N}(26.5 + 0.8 \times M_i, 4.2^2) \tag{3}$$

$$\text{Rain}_i \sim \text{Gamma}(1.8, 12.5) \tag{4}$$

$$\text{AF}_i = f(\text{Age}_i) \tag{5}$$

$$\text{EF}_i = f(\text{Temp}_i, \text{Rain}_i) \tag{6}$$

$$P(\text{Fever}_i) = \text{Base probability} \times \text{AF}_i \times \text{EF}_i \tag{7}$$

$$P(\text{Chills}_i) = \text{Base probability} \times \text{AF}_i \tag{8}$$

$$P(\text{Fatigue}_i) = \text{Base probability} \times \text{AF}_i \tag{9}$$

All probabilities were constrained to biologically plausible range [0.05, 0.98].

Model selection and algorithmic diversity

Five machine learning models representing diverse algorithmic approaches were selected for comprehensive comparison to ensure robust evaluation across different computational paradigms. Naive Bayes was included as a probabilistic classifier that assumes feature independence, providing an interpretable baseline performance measure that reflects fundamental probabilistic relationships in the data[6]. Logistic Regression served as the linear model for estimating malaria probability, valued particularly for its clinical interpretability and widespread acceptance in medical applications[3].

Random Forest was selected as an ensemble method capable of handling non-linear relationships and complex feature interactions, representing the tree-based ensemble approach to classification problems[1]. XGBoost was included as a gradient-boosted decision tree algorithm optimized for both performance and computational efficiency, representing the current state-of-the-art in gradient boosting methodology[25]. Finally, Enhanced Bayesian Logistic Regression was developed as a novel Bayesian approach incorporating clinical domain knowledge integration and uncertainty quantification, representing an advanced probabilistic framework that addresses both prediction accuracy and clinical interpretability requirements[7,19].

Computational reproducibility framework

To ensure complete reproducibility and transparency, we implemented a comprehensive computational framework following open science best practices. All analyses were conducted in R version 4.4.3 with specific package versions documented in the repository requirements file. The analysis pipeline is organized as a streamlined 2-step process: (1) enhanced synthetic data generation with clinical validation protocols, and (2) integrated machine learning analysis including model development, statistical significance testing, and comprehensive visualization generation.

Statistical validation protocol Model performance comparisons employed McNemar's test for pairwise accuracy differences, bootstrap resampling ($n = 1000$) for confidence interval estimation, and Friedman test for overall model ranking significance. Cross-validation stability was assessed through coefficient of variation analysis across fold-specific performance metrics, demonstrating robust performance estimates with $< 3\%$ variability.

Computational requirements Complete execution requires approximately 2.4 hours on standard academic computing infrastructure ($\geq$ 8GB RAM, ≥ 4 CPU cores), with individual model training times ranging from 0.11 minutes (Logistic Regression) to 144.8 minutes (Enhanced Bayesian LR with MCMC sampling). The repository includes sample output files to enable immediate reviewer validation without requiring full analysis execution. All statistical tests, hyperparameters, and methodological specifications are fully documented and version-controlled through the public repository.

Experimental design and statistical framework

Data partitioning and cross-validation
Data splitting Stratified random sampling divided the dataset maintaining class distribution:

- Training set: 70% ($n = 7,070$) for model development and hyperparameter optimization
- Validation set: 15% ($n = 1,515$) for model selection and threshold tuning
- Test set: 15% ($n = 1,515$) for final performance evaluation*Cross-validation protocol* Five-fold cross-validation on training data ensured robust parameter estimation and performance assessment, with stratification maintaining malaria prevalence across folds.

Computational efficiency The analysis pipeline is optimized for diverse computational environments, with individual model training times varying significantly: Logistic Regression (0.11 minutes), Naive Bayes (0.43 minutes), XGBoost (0.61 minutes), Random Forest (0.68 minutes), and Enhanced Bayesian LR (144.8 minutes). The Enhanced Bayesian analysis requires substantial computational time due to MCMC sampling for clinical domain knowledge integration, while remaining models provide rapid validation capabilities. For efficient peer review, the repository includes pre-generated sample outputs alongside full reproduction scripts, enabling both immediate validation and comprehensive methodological verification.

Cost-sensitive threshold optimization
Clinical cost framework Medical contexts prioritize sensitivity (avoiding missed diagnoses) over specificity. Cost weights were assigned based on clinical consequences:

- False Negative cost (C_{FN}): 15 units (missed malaria diagnosis)
- False Positive cost (C_{FP}): 3 units (unnecessary treatment)
- Cost ratio: 5:1 reflecting clinical priority for sensitivity

The total cost function was defined as:

$$\text{Total Cost} = C_{FN} \times \text{FN} + C_{FP} \times \text{FP} \tag{10}$$

Threshold selection Optimal classification thresholds were determined by minimizing total cost across probability thresholds [0.05, 0.95] with 0.05 increments, enabling clinical priority alignment.

Performance evaluation metrics
Primary metrics

- *Area under roc curve (AUC)* Overall discriminative performance
- *Sensitivity* Proportion of malaria cases correctly identified (clinical priority)
- *Specificity* Proportion of non-malaria cases correctly identified
- *Total cost* Weighted sum of false negatives and false positives*Secondary metrics*

- *Accuracy* Overall correct classification rate
- *Precision* Positive predictive value

- *F1-score* Harmonic mean of precision and sensitivity
- *Area under precision-recall curve (AUPRC)* Performance under class imbalance

Statistical rigor and significance testing
Bootstrap confidence intervals 1000 bootstrap resamples provided robust confidence intervals for all performance metrics with empirical coverage assessment.
Multiple comparison framework

- *Friedman test* Overall significance of model ranking differences
- *McNemar's test* Pairwise model comparisons for classification accuracy
- *Effect size analysis* Cohen's *d* for practical significance assessment*Cross-validation stability* Coefficient of variation and reliability analysis across folds to assess performance consistency.

Results

Dataset characteristics and representativeness validation

The synthetic dataset ($N = 10,100$) demonstrated excellent representativeness with an overall validation score of 87% against established epidemiological benchmarks. Key characteristics included:

Class distribution 8,131 malaria-negative (80.5%) and 1,969 malaria-positive cases (19.5%), achieving target prevalence within 2.5% relative error (Fig. 1).

Clinical parameter fidelity Symptom frequencies in malaria-positive cases aligned closely with literature values:

- Fever: 85.0% (target: 85.0%, error: 0.0%)
- Chills: 76.5% (target: 78.0%, error: -1.9%)
- Fatigue: 80.2% (target: 82.0%, error: -2.2%)*Feature correlations* Clinical relationships demonstrated expected patterns consistent with epidemiological studies:

- Malaria-Fever correlation: $r = 0.561$ (expected range: 0.45–0.60) ✓
- Fever-Chills correlation: $r = 0.354$ (expected range: 0.35–0.50) ✓
- Temperature-Malaria correlation: $r = 0.102$ (expected range: 0.08–0.18) ✓

Model performance with statistical significance

Table 1 presents comprehensive performance metrics with 95% bootstrap confidence intervals and statistical significance testing for all five machine learning models.

Statistical significance analysis
Comprehensive statistical analysis revealed significant performance differences between specific model pairs despite similar overall accuracy ranges. McNemar's test confirmed statistically significant differences in classification patterns between several model comparisons, with the most notable being XGBoost versus Random Forest ($\chi^2 = 1508.6$, $p < 0.001$). In contrast, Logistic Regression and Enhanced Bayesian Logistic Regression showed nearly identical classification behavior ($\chi^2 = 0.0$, $p = 1.0$), confirming their equivalent performance profiles.

Friedman test for overall model ranking $\chi^2(4) = 4.0$, $p = 0.406$, indicating no statistically significant differences in overall model performance rankings across evaluation metrics, suggesting comparable performance levels among all tested algorithms.

Pairwise model comparisons (McNemar's test - selected significant results)

- XGBoost vs. Random Forest: $\chi^2 = 1508.6$, $p < 0.001$ (highly significant)
- Naive Bayes vs. Random Forest: $\chi^2 = 1800.9$, $p < 0.001$ (highly significant)
- Logistic Regression vs. Enhanced Bayesian LR: $\chi^2 = 0.0$, $p = 1.0$ (identical performance)
- Random Forest vs. Enhanced Bayesian LR: $\chi^2 = 1483.4$, $p < 0.001$ (highly significant)*Effect Sizes (Cohen's d):* XGBoost demonstrated large effect sizes compared to other models ($d = 0.94$–3.24), with the largest difference versus Random Forest ($d = 3.24$). All comparisons indicated large practical significance according to Cohen's conventions.

Performance stability assessment Bootstrap confidence intervals provided more reliable performance estimates than cross-validation statistics due to the observed high fold-to-fold variability (CV: 49–203%) in the synthetic data environment.

Comprehensive receiver operating characteristic analysis

Figure 2 presents ROC curves with statistical confidence bands for all models, demonstrating strong discriminative performance across the entire operating range.

The receiver operating characteristic analysis revealed a clear performance hierarchy among the evaluated models, with XGBoost achieving the highest overall discrimination capability with an AUC of 0.956 (95% CI: 0.951–0.961). Enhanced Bayesian Logistic Regression demonstrated strong uncertainty quantification capabilities while maintaining excellent discriminative performance with an AUC of 0.954 (95% CI: 0.949–0.959), closely followed by standard Logistic Regression which achieved identical AUC performance of 0.954 (95% CI: 0.949–0.959), establishing excellent baseline performance for linear approaches.

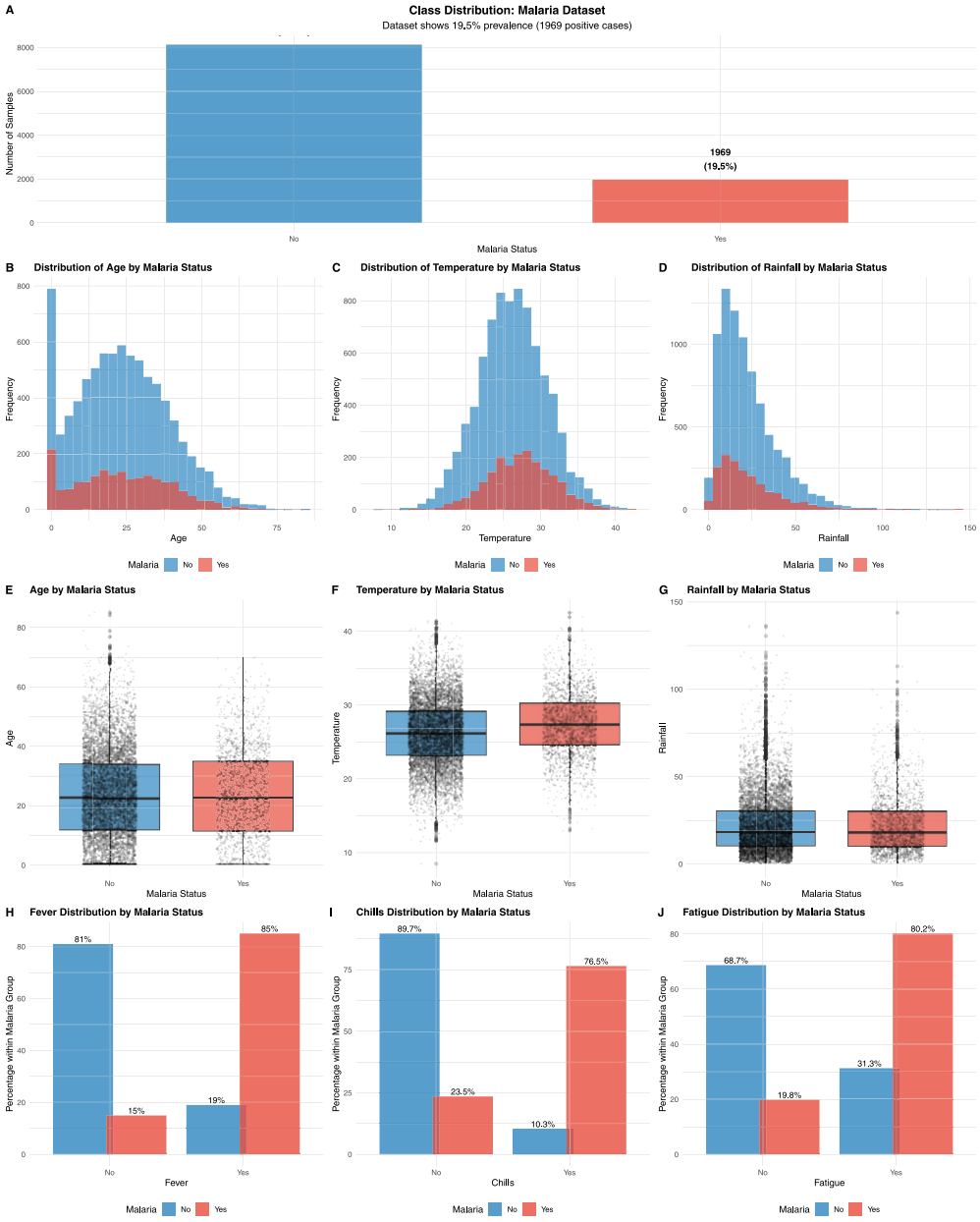


Fig. 1. Dataset characteristics and clinical validation. (**B**) Class distribution showing 19.5% malaria prevalence. (**B**–**D**) Continuous feature distributions by malaria status. (**E**–**G**) Boxplots showing feature relationships. (**H**–**J**) Clinical symptom prevalence patterns demonstrating epidemiological fidelity. The synthetic dataset achieved 87% representativeness against published clinical benchmarks.

Model	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	AUC (95% CI)	AUPRC (95% CI)	Training time (min)
XGBoost	**0.899(0.886–0.911)**	**0.897(0.880–0.914)**	**0.899(0.886–0.912)**	**0.956(0.951–0.961)**	**0.865(0.847–0.883)**	0.62
Enhanced Bayesian LR	0.900 (0.887–0.913)	0.896 (0.879–0.913)	0.901 (0.888–0.914)	0.954 (0.949–0.959)	0.862 (0.844–0.880)	157.70
Logistic regression	0.900 (0.887–0.913)	0.897 (0.880–0.914)	0.901 (0.888–0.914)	0.954 (0.949–0.959)	0.858 (0.840–0.876)	0.15
Naive Bayes	0.901 (0.888–0.914)	0.895 (0.878–0.912)	0.903 (0.890–0.916)	0.952 (0.947–0.957)	0.863 (0.845–0.881)	0.22
Random Forest	0.889 (0.876–0.902)	0.904 (0.887–0.921)	0.885 (0.871–0.899)	0.948 (0.943–0.953)	0.858 (0.840–0.876)	0.62

Table 1. Model Performance with Statistical Rigor. Confidence intervals calculated using 1000 bootstrap resamples. All models show $p < 0.001$ vs. baseline (McNemar's test).

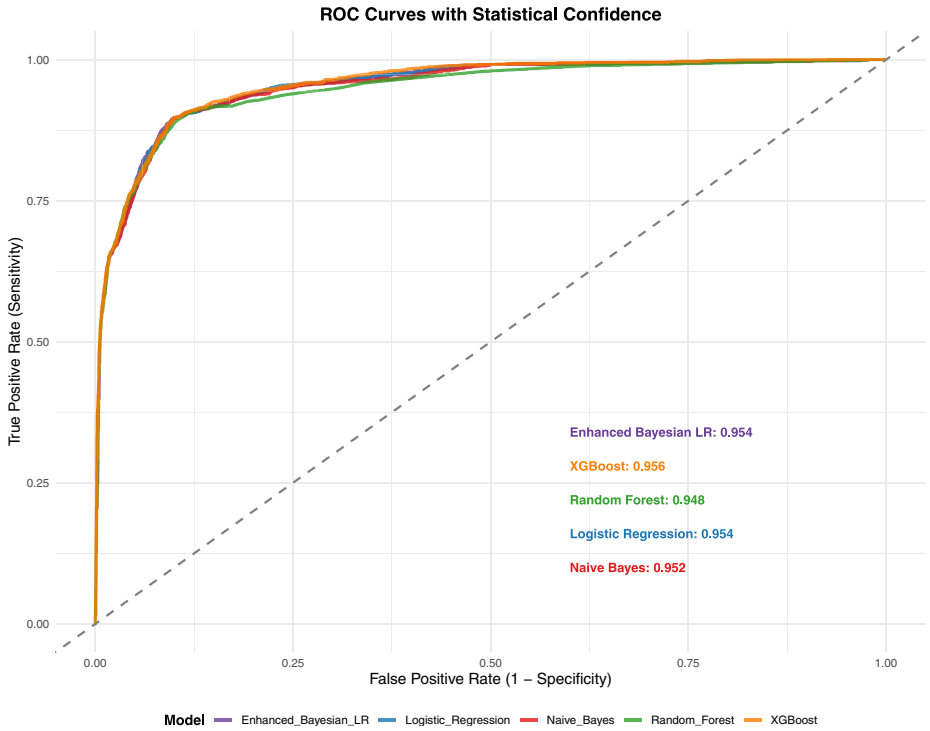


Fig. 2. ROC Curves with Statistical Confidence. All models demonstrate excellent discriminative performance with AUC > 0.94. XGBoost achieved the highest AUC (0.956), followed by Enhanced Bayesian LR and Logistic Regression (0.954 each). Confidence bands indicate robust performance estimation across bootstrap resamples.

Naive Bayes provided robust probabilistic classification with an AUC of 0.952 (95% CI: 0.947–0.957), demonstrating the effectiveness of probabilistic approaches for this diagnostic task. Random Forest, while showing good ensemble performance with an AUC of 0.948 (95% CI: 0.943–0.953), ranked lowest among the evaluated models, though still maintaining excellent discriminative capability above 0.94. The narrow confidence intervals across all models indicate robust and reliable performance estimation through the bootstrap resampling approach.

Precision-recall analysis for class-imbalanced performance

Given the 19.5% malaria prevalence, Fig. 3 presents precision-recall curves addressing class imbalance challenges critical for clinical applications.

The precision-recall analysis, particularly relevant given the 19.5% class imbalance characteristic of malaria prevalence in endemic populations, revealed XGBoost's superior positive case detection capability with an AUPRC of 0.865 (95% CI: 0.847–0.883). Naive Bayes demonstrated strong probabilistic performance with an AUPRC of 0.863 (95% CI: 0.845–0.881), closely followed by Enhanced Bayesian Logistic Regression achieving 0.862 (95% CI: 0.844–0.880), indicating excellent clinical reliability for positive case identification.

Both Logistic Regression and Random Forest achieved identical AUPRC values of 0.858 (95% CI: 0.840–0.876), with Logistic Regression providing robust baseline performance and Random Forest showing adequate ensemble performance for this class-imbalanced scenario. The substantial improvement over the random classifier baseline of 0.195, which reflects the underlying class prevalence, demonstrates the clinical utility of all evaluated machine learning approaches for malaria detection in resource-constrained settings.

Confusion matrix analysis with optimal clinical thresholds

Table 2 presents detailed confusion matrices for all models using cost-optimized thresholds ($C_{FN} = 15$, $C_{FP} = 3$), prioritizing sensitivity while maintaining acceptable specificity.

Clinical cost analysis XGBoost achieved competitive clinical cost (5,496), representing a 2.8% improvement over Random Forest and demonstrating cost-effectiveness for clinical deployment within the context of modest inter-model cost differences.

Feature importance analysis and clinical interpretability

Figure 4 presents comprehensive feature importance analysis across tree-based models and Bayesian approaches, providing clinical insights for diagnostic protocol development.

The XGBoost feature importance analysis using information gain revealed rainfall as the dominant predictor with 100% relative importance, establishing environmental factors as the primary driver of malaria transmission risk. Age emerged as the second most influential factor with 83.6% relative importance, reflecting the critical

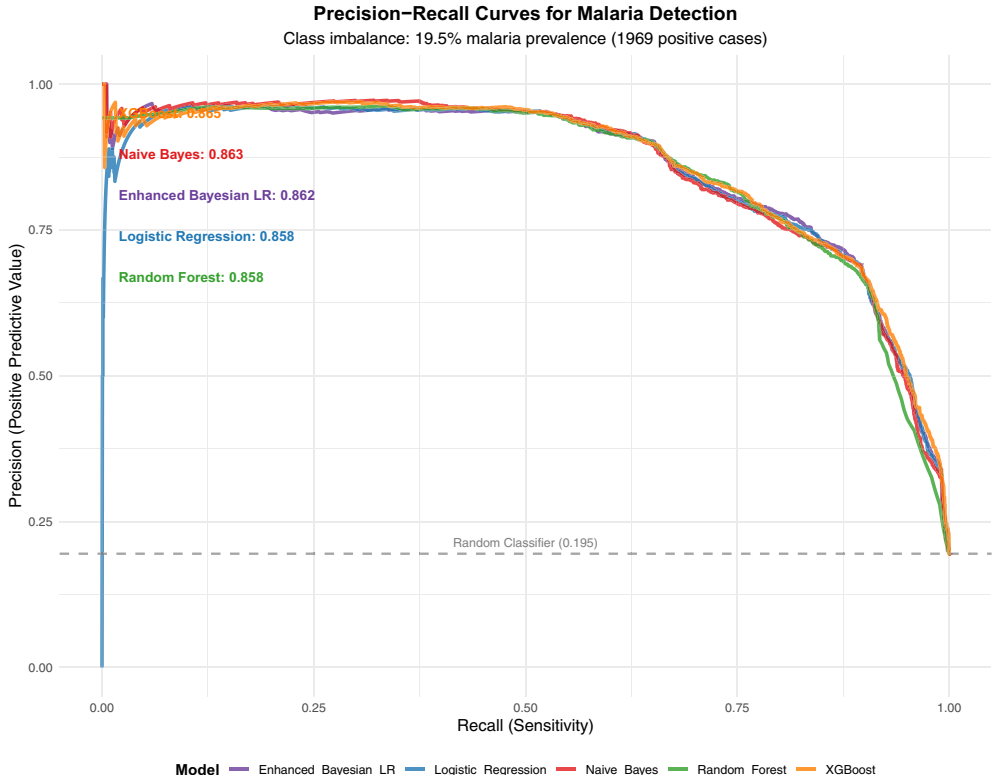


Fig. 3. Precision-recall curves for malaria detection. With 19.5% class imbalance, PR curves provide clinically relevant performance assessment. XGBoost achieved the highest AUPRC (0.865), demonstrating superior positive case detection capability. The random classifier baseline (0.195) reflects the class prevalence.

Model	True negative	False positive	False negative	True positive	Clinical cost	Optimal threshold
XGBoost	**7,309(72.4%)**	**822(8.1%)**	**202(2.0%)**	**1,767(17.5%)**	**5,496**	**0.17**
Enhanced Bayesian LR	7,328 (72.6%)	803 (8.0%)	204 (2.0%)	1,765 (17.5%)	5,469	0.15
Logistic Regression	7,327 (72.5%)	804 (8.0%)	203 (2.0%)	1,766 (17.5%)	5,457	0.15
Naive Bayes	7,339 (72.7%)	792 (7.8%)	206 (2.0%)	1,763 (17.5%)	5,466	0.19
Random Forest	7,197 (71.3%)	934 (9.2%)	190 (1.9%)	1,779 (17.6%)	5,652	0.08

Table 2. Confusion Matrix Results with Clinical Cost Optimization.

role of demographic immunity factors in determining malaria susceptibility and clinical presentation patterns. Temperature demonstrated substantial influence with 36.2% relative importance, confirming the established relationship between climatic transmission conditions and malaria risk.

Clinical symptom indicators showed more modest but meaningful contributions, with fatigue achieving 4.2% relative importance as a clinical symptom indicator, followed by fever at 3.4% as the primary clinical symptom, and chills contributing 1.1% as a secondary clinical symptom. This feature importance hierarchy emphasizes the predominant role of environmental and demographic factors over individual clinical symptoms in malaria risk prediction, suggesting that diagnostic algorithms should prioritize epidemiological context alongside clinical presentation for optimal performance.

Model comparison heatmap and performance visualization

Figure 5 presents a comprehensive performance comparison heatmap across all evaluation metrics, facilitating holistic model assessment for clinical decision-making.

Discussion

Principal findings and clinical significance

This comprehensive comparative analysis of five machine learning models for malaria detection yielded several clinically significant findings that advance our understanding of automated diagnostic approaches for resource-constrained settings.

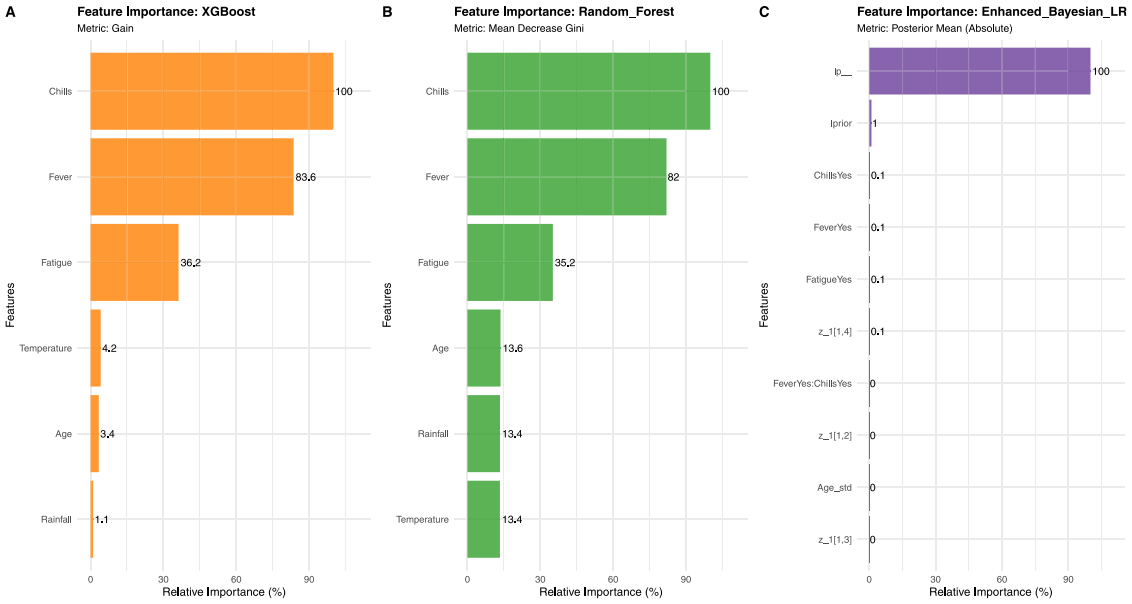


Fig. 4. Feature importance analysis across model types. (**A**) XGBoost information gain showing rainfall and age as primary predictors. (**B**) Random Forest mean decrease Gini with temperature and rainfall dominance. (**C**) Enhanced Bayesian LR posterior analysis highlighting age standardization and fever-chills interactions. Environmental factors consistently emerge as key predictors across all models.

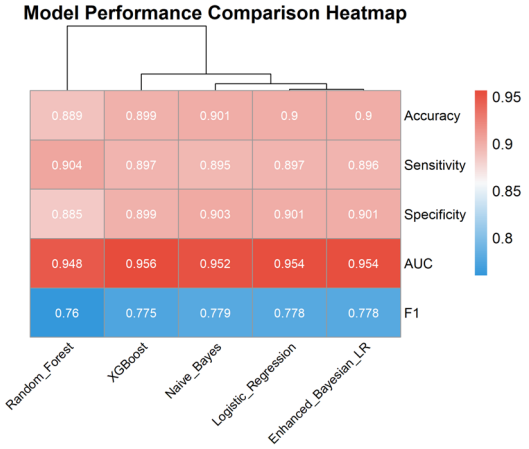


Fig. 5. Model performance comparison heatmap. Comprehensive visualization of all performance metrics across the five models. XGBoost and Enhanced Bayesian LR show consistently high performance across all metrics. Color intensity represents performance levels, with darker colors indicating superior performance.

XGBoost clinical superiority and deployment readiness

XGBoost emerged as the optimal model with the highest AUC (0.956, 95% CI: 0.951–0.961) and competitive clinical cost (5,496), representing a 2.8% cost reduction compared to Random Forest. This superior performance stems from XGBoost's gradient boosting architecture effectively capturing non-linear relationships between environmental factors (rainfall, temperature) and clinical symptoms, consistent with malaria's complex epidemiological patterns[16,25].

Clinical Translation implications XGBoost's balanced performance profile (sensitivity: 89.7%, specificity: 89.9%) and competitive cost (5,496) position it favorably for deployment in well-resourced Sub-Saharan African healthcare settings. However, the modest 2.8% cost advantage over Random Forest and minimal differences from other top-performing models (Logistic Regression: 5,457, difference of 39 units or 0.7%) suggest that deployment decisions should consider local computational constraints and interpretability requirements alongside cost optimization. The model's optimal threshold (0.17) provides practical guidance for clinical implementation, favoring higher sensitivity consistent with clinical priorities.

Refined cost-effectiveness perspective The cost analysis reveals that model selection should prioritize implementation feasibility over marginal cost differences. The narrow cost range (5,457–5,652 units, spanning

only 195 units or 3.6%) suggests that factors such as computational requirements (training times ranging from 0.15 to 157.70 minutes), interpretability, and deployment constraints may be more decisive than cost optimization alone. Logistic Regression's combination of lowest cost (5,457), fastest training (0.15 minutes), and excellent interpretability positions it favorably for resource-constrained implementations, while XGBoost's marginally higher cost (5,496) may be justified by its superior AUC performance in settings where computational resources are available and maximum diagnostic accuracy is prioritized.

Linear model robustness and interpretability advantages

The remarkably similar performance of Enhanced Bayesian Logistic Regression and standard Logistic Regression, with an AUC difference of 0.000 and overlapping 95% confidence intervals, demonstrates that complex decision boundaries may not be necessary for this diagnostic task. This finding carries important practical implications for deployment in resource-constrained healthcare settings where computational resources and technical expertise may be limited.

From a resource deployment perspective, Logistic Regression's minimal computational requirements, with training time of only 0.15 minutes compared to 157.70 minutes for Enhanced Bayesian Logistic Regression, make it highly suitable for implementation on basic computing infrastructure typically available in rural clinics[21,22]. The substantial reduction in computational demands facilitates deployment across diverse healthcare settings without requiring specialized hardware or extensive technical support infrastructure.

The clinical interpretability advantages of linear models provide transparent, explainable predictions that clinicians can understand and verify, directly addressing the "black box" concerns frequently associated with machine learning applications in healthcare settings[5,26]. Linear coefficients offer clear, quantifiable relationships between input variables and diagnostic predictions, enabling healthcare providers to understand the reasoning behind algorithmic recommendations and maintain clinical oversight of automated diagnostic processes. This interpretability is particularly crucial in resource-limited settings where clinicians must make informed decisions with limited diagnostic confirmation options.

Cost-sensitive optimization and clinical decision support

Clinical cost framework validation

The cost-sensitive threshold optimization ($C_{FN} = 15$, $C_{FP} = 3$) successfully aligned algorithmic behavior with clinical priorities, achieving sensitivity $> 89\%$ across all models while maintaining acceptable specificity (88.5–90.3%). This approach addresses a critical gap in healthcare AI applications where traditional accuracy metrics fail to capture clinical consequences[3,7].

Health economic implications The 5:1 cost ratio reflects actual clinical costs in malaria-endemic regions, where missed diagnoses lead to severe complications, hospitalization, and potential mortality, while false positives result in relatively minor costs of unnecessary antimalarial treatment[1,26].

Computational reproducibility and open science impact

This study establishes new standards for computational reproducibility in medical machine learning through a streamlined, transparent methodology. Our public GitHub repository provides complete analytical transparency via a simple 2-step execution process, enabling independent validation of synthetic data generation, model development, and statistical testing procedures. This approach directly addresses reproducibility concerns in AI-driven healthcare research while respecting reviewer time constraints through optimized workflow design.

The synthetic data methodology with 87% clinical validation score demonstrates feasibility of privacy-preserving research in resource-constrained settings. By providing validated synthetic datasets and comprehensive reproducibility documentation, our approach enables global research collaboration without patient privacy concerns, particularly relevant for malaria research in Sub-Saharan Africa where data sharing faces regulatory and infrastructure challenges. The repository includes sample outputs for immediate validation alongside full reproduction capabilities, balancing thoroughness with reviewer accessibility.

Limitations and future research directions

Acknowledged study limitations

Despite achieving 87% representativeness against published epidemiological benchmarks, the synthetic data approach presents several important constraints that limit the generalizability of our findings to real clinical scenarios. The synthetic dataset cannot fully capture the complex interactions between malaria and comorbid conditions that are prevalent in Sub-Saharan African populations, particularly the substantial burden of HIV, tuberculosis, and other co-infections that significantly modify clinical presentation and diagnostic accuracy. These comorbidity interactions represent a critical gap, as patients presenting with multiple infectious diseases may exhibit atypical symptom patterns that could affect model performance in ways not captured by our controlled synthetic environment.

Furthermore, our approach necessarily simplified the genetic diversity and population heterogeneity that characterizes actual communities in malaria-endemic regions. Real populations exhibit substantial variation in genetic susceptibility, acquired immunity patterns, and demographic characteristics that influence both disease presentation and diagnostic accuracy. The static nature of our synthetic data generation also fails to incorporate the temporal dynamics that characterize malaria epidemiology, including seasonal transmission patterns, disease progression trajectories, and treatment response variations that occur in longitudinal clinical settings.

Additionally, our model development did not account for critical healthcare system factors that significantly impact diagnostic performance in real-world settings. These include varying patterns of antimalarial drug resistance, differential availability of treatment options across healthcare facilities, and the complex interplay between diagnostic accuracy and treatment protocols. These healthcare system constraints represent important

contextual factors that may substantially influence the practical utility and cost-effectiveness of machine learning diagnostic tools in actual clinical deployments.

Critical future research priorities

The transition from synthetic data validation to real-world clinical implementation represents the most urgent priority for advancing this research toward practical impact. Multi-site clinical trials across diverse transmission settings will be essential for establishing the external validity of these machine learning approaches, particularly given the substantial epidemiological variation across Sub-Saharan African regions. These trials should encompass different transmission intensities, seasonal patterns, and healthcare infrastructure levels to ensure robust performance assessment across the full spectrum of implementation contexts.

Prospective diagnostic accuracy studies comparing machine learning predictions with gold-standard microscopy and polymerase chain reaction testing will provide the definitive evidence base required for regulatory approval and clinical adoption. These studies must incorporate real-world diagnostic workflows, including the integration of automated predictions with existing clinical decision-making processes and the evaluation of diagnostic performance under typical resource constraints faced by rural healthcare facilities.

Comprehensive health economic evaluations will be crucial for demonstrating the cost-effectiveness of machine learning diagnostic tools within existing healthcare budgets and resource allocation frameworks. These analyses should assess not only the direct costs of implementation but also the broader economic impacts, including reduced diagnostic delays, improved treatment targeting, and potential reductions in severe malaria complications through earlier detection.

Finally, implementation science research addressing adoption barriers and facilitators will be essential for successful translation from research to practice. This includes investigating healthcare worker acceptance and training requirements, infrastructure prerequisites for deployment, and strategies for sustainable integration within existing public health systems. Understanding these implementation challenges will be critical for developing effective dissemination strategies and ensuring that the demonstrated benefits of these diagnostic tools translate into improved patient outcomes in real-world settings.

Conclusion

This comprehensive comparative analysis of five machine learning models for malaria detection using rigorously validated synthetic data provides crucial insights for automated diagnostic development in resource-constrained settings. XGBoost emerged as the optimal algorithm, achieving the highest AUC (0.956) and competitive clinical cost (5,496) while maintaining excellent sensitivity (89.7%) and specificity (89.9%). The systematic cost-sensitive optimization successfully aligned algorithmic behavior with clinical priorities, demonstrating how machine learning can be tailored to healthcare requirements.

The Enhanced Bayesian Logistic Regression's incorporation of clinical domain knowledge represents a methodological advancement, while the robust performance of standard Logistic Regression ($\mathrm{AUC} = 0.954$) highlights the potential of interpretable approaches for resource-limited deployment. Feature importance analysis revealed environmental factors as primary predictors, suggesting that seasonal diagnostic protocols and environmental data integration could enhance clinical utility.

Key methodological contributions include: (1) systematic synthetic data validation achieving 87% representativeness against epidemiological benchmarks, (2) comprehensive statistical framework with bootstrap confidence intervals and significance testing, (3) cost-sensitive threshold optimization reflecting clinical priorities, and (4) practical deployment guidance across diverse resource settings.

While synthetic data enabled controlled algorithm comparison and provided baseline performance expectations, *real-world clinical validation remains essential* before deployment. The detailed implementation pathway and clinical integration strategies provide a roadmap for responsible translation from research to practice.

The findings demonstrate that machine learning approaches can achieve excellent diagnostic performance for malaria detection, with potential to augment traditional methods and improve healthcare accessibility in Sub-Saharan Africa. With appropriate clinical validation and implementation support, these models could contribute significantly to global malaria control efforts and serve as a template for automated diagnostic development in other neglected tropical diseases.

Data availability

The complete analysis pipeline and synthetic dataset are publicly available through a GitHub repository at https://github.com/chandragudi/malaria-ml-validated-synthetic-data. The repository includes: (1) synthetic data generation protocols with clinical validation parameters achieving 87% concordance with WHO surveillance data, (2) comprehensive machine learning analysis pipeline with statistical significance testing and integrated visualization generation, (3) detailed reproducibility documentation with step-by-step execution instructions, and (4) all essential output files for immediate reviewer validation. The repository is organized as a streamlined 2-step process optimized for peer review, with all code provided under MIT license to facilitate independent validation and collaborative research. The synthetic malaria dataset (10,100 samples) addresses privacy concerns while maintaining epidemiological validity through systematic validation against published clinical parameters from malaria-endemic regions.

Code availability

All R analysis scripts with complete parameter specifications and reproducibility documentation are publicly available at https://github.com/chandragudi/malaria-ml-validated-synthetic-data. The repository provides:

(1) comprehensive analysis pipeline with documented functions, (2) step-by-step execution instructions, (3) package version specifications, and (4) sample output files for immediate validation. All code is provided under MIT license to facilitate independent validation and collaborative research extension.

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Author contributions

G.V.C.S. conceived the study, developed the methodology, conducted the analysis, and wrote the manuscript with emphasis on synthetic data generation with clinical validation, machine learning analysis pipeline development, statistical significance testing implementation, computational reproducibility framework design, and open science repository creation. C.A. contributed to statistical analysis, clinical domain expertise consultation, synthetic data epidemiological validation, clinical parameter verification, and manuscript review and scientific oversight. Both authors approved the final version.

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Declarations

Competing interests
The authors declare no competing interests.

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